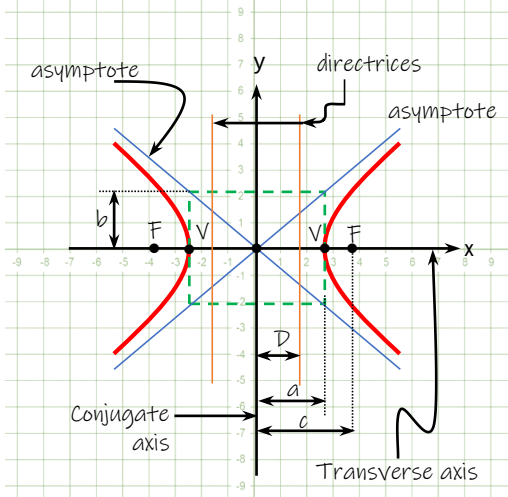




## Hyperbola

A hyperbola is a locus of a point which moves so that the difference of the distances to the fixed points (foci) is constant and is equal to the length of the transverse axis (2a).



### General Equation

A. Transverse axis horizontal

$$Ax^2 - Cy^2 + Dx + Ey + F = 0$$

**Transverse axis** is the axis that passes through the foci, vertices and the center of the hyperbola while the **conjugate axis** is the one that is perpendicular to the transverse axis.

$$\text{Length of the transverse axis} = 2a = 2\sqrt{C}$$

$$\text{Length of the conjugate axis} = 2b = 2\sqrt{A}$$

where: A and C are the numerical coefficients (absolute value) of  $x^2$  and  $y^2$ , respectively.

Also, the relationship between a, b and c is

$$a^2 + b^2 = c^2$$

B. Transverse axis vertical

$$Cx^2 - Ay^2 + Dx + Ey + F = 0$$

### Standard Equation: C(h,k)

Transverse axis horizontal

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Transverse axis vertical

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

### Standard Equation: C(h,k)

Transverse axis horizontal

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Transverse axis vertical

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

### LATUS RECTUM

$$LR = \frac{2b^2}{a}$$

### ECCENTRICITY

$$e = \frac{a}{D} = \frac{c}{a} > 1.0$$

where: a = semi major axis & b = semi minor axis

When the equation given is a general equation rather than standard equation, the center (h,k) of a hyperbola can be calculated by converting the general equation to standard using the process known as completing the square. The following formulas can be obtained:

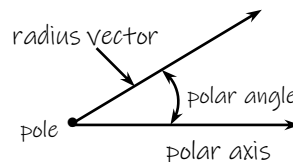
$$h = \frac{-D}{2A}$$

$$k = \frac{-E}{2C}$$

## Polar Coordinates

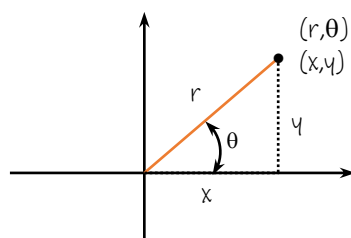
Polar coordinates (r,θ) refers to the coordinates of a point in a system of coordinates where the position of a point is determined by the length of ray segment (the radius vector) from a fixed origin (the pole) and the angle (the polar angle) the ray (the vector) makes with a fixed line (the polar axis).

Polar angle is sometimes called the vectorial angle, the argument, the amplitude, or the azimuth of a point.



## Polar ↔ Rectangular Coordinates

Relationship between polar coordinates and rectangular coordinates:



$$x = r \cos \theta$$

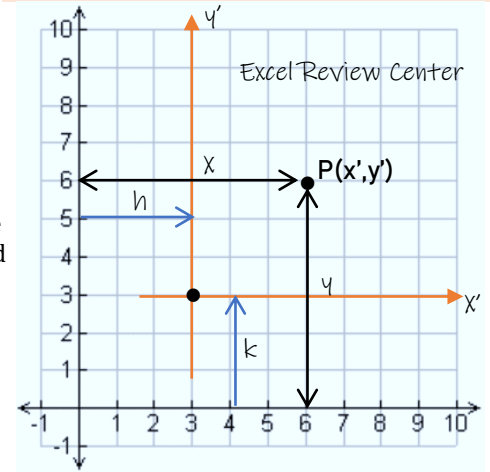
$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan \frac{y}{x}$$

## Translation of Axes

is a mapping from an xy-Cartesian coordinate system to an x'y'-Cartesian coordinate system in which the x' axis is parallel to the x axis and k units away, and the y' axis is parallel to the y axis and h units away.

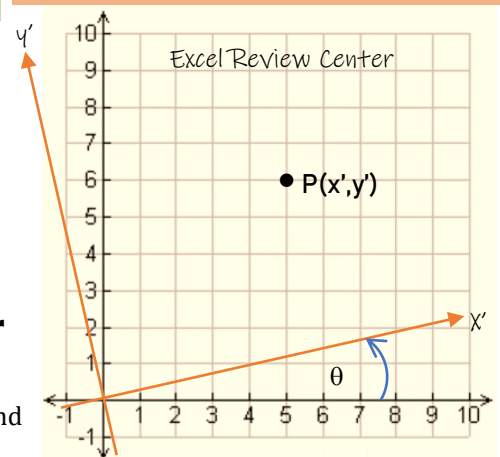


$$x' = x - h$$

$$y' = y - k$$

## Rotation of Axes

is a mapping from an xy-Cartesian coordinate system to an x'y'-Cartesian coordinate system in which the origin is kept fixed and the x' and y' axes are obtained by rotating the x and y axes counterclockwise through an angle θ.



$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$